

A Moser-Lattice and Dijkstra Solver for Reflection and Refraction

From one mandatory surface to a finite triangular prism

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Abstract

We validate one reusable graph-based geometrical-optics solver on four benchmarks: reflection from a plane mirror, refraction across a horizontal interface, refraction across a non-grid-aligned straight interface, and an ordered air–glass–air path through a finite triangular prism. Optical paths are discretized by a Moser lattice, graph edges carry optical-path-length weights, and two Dijkstra distance fields are combined on a mandatory surface. For a tilted discontinuity, every intersected main cell is split into two conforming subcells and receives explicit interface nodes; the shortest-path engine itself is unchanged. The prism introduces two unknown contacts. One entry distance field initializes a single multi-source glass propagation, which is then combined with a backward exit field. Convex material-region masks enforce one entry, one exit, and no base contact. Analytic reflection and continuous Snell solutions are external benchmarks only. At $h = 1$ and $q = 7$, all four baseline relative optical-path errors remain below 0.1%; the prism error is 5.84×10^{-4} . Resolution sweeps and angle–phase ensembles show that increasing q enriches the available directions, whereas decreasing h alone does not necessarily remove directional lattice bias. A small functional error remains insufficient evidence of an accurate ray.

1 Motivation and scope

Fermat’s principle states that a physical optical path is stationary with respect to nearby admissible paths. For a ray γ in a scalar refractive index field $n(x, y)$, the optical functional is

$$\mathcal{L}[\gamma] = \int_{\gamma} n(x, y) \, ds. \quad (1)$$

Graph methods with positive edge weights naturally recover global minima of a discrete version of Eq. (1). Arunthong et al. demonstrated the use of Moser lattices and Dijkstra’s algorithm for several variational problems, including ray tracing and Snell refraction [1]. The present project adopts that common numerical foundation and develops a controlled optical research program around constraints, convergence, curved interfaces, image recovery, and aberrations.

This report covers four controlled benchmarks: one reflection from a horizontal plane mirror, transmission across a horizontal interface, and transmission across a straight interface that cuts the Cartesian cells at an arbitrary angle, followed by transmission through two finite faces of a triangular prism. Its purpose is not to rediscover the laws of reflection or refraction. Instead, it tests whether one universal Moser-lattice and Dijkstra engine can recover all four rays while exposing discretization error honestly. Analytic optics is used strictly for validation.

2 Continuous plane-mirror benchmark

Let the source and receiver be

$$A = (x_A, y_A), \quad B = (x_B, y_B), \quad y_A, y_B > 0, \quad (2)$$

above the mirror $\Gamma = \{(x, 0) : x \in \mathbb{R}\}$. A path required to touch the mirror once at $P = (x_P, 0)$ has optical length

$$\mathcal{L}(P) = n \left[\sqrt{(x_P - x_A)^2 + y_A^2} + \sqrt{(x_B - x_P)^2 + y_B^2} \right]. \quad (3)$$

For a homogeneous medium, reflecting B across the mirror gives $B' = (x_B, -y_B)$. The straight segment AB' intersects the mirror at

$$x_{P,\text{exact}} = \frac{x_A y_B + x_B y_A}{y_A + y_B}. \quad (4)$$

This point minimizes Eq. (3) and satisfies $\theta_i = \theta_r$. In the baseline geometry,

$$A = (-3, 2), \quad B = (4, 3), \quad n = 1, \quad (5)$$

so that

$$P_{\text{exact}} = (-0.2, 0), \quad \mathcal{L}_{\text{exact}} = \sqrt{74} = 8.602325267 \dots \quad (6)$$

3 Universal graph formulation

3.1 Moser lattice

The rectangular domain is divided into square main cells of side length h . Each cell side contains q interior Moser points and therefore $q + 1$ equal subintervals. Within every main cell, all pairs of boundary nodes are connected by straight chords. Geometrically identical edges shared by neighboring cells are deduplicated globally. Source and receiver points need not coincide with lattice nodes: an off-grid endpoint is inserted into every cell containing it and connected by the same complete-within-cell rule.

The two resolution parameters play different roles. Decreasing h shortens the spatial steps, whereas increasing q enriches the locally available directions. Keeping these parameters separate is essential for diagnosing directional lattice bias.

3.2 Optical edge weights and shortest paths

Every graph edge e_{ij} has the positive weight

$$w_{ij} = \int_{e_{ij}} n(x, y) \, ds. \quad (7)$$

For the present homogeneous benchmark this integral is exact:

$$w_{ij} = n \|\mathbf{r}_j - \mathbf{r}_i\|_2 > 0. \quad (8)$$

The same weight interface is retained for future media. Smooth index fields can be integrated by quadrature; discontinuous interfaces must be intersected and split exactly rather than averaged across an unsplit edge.

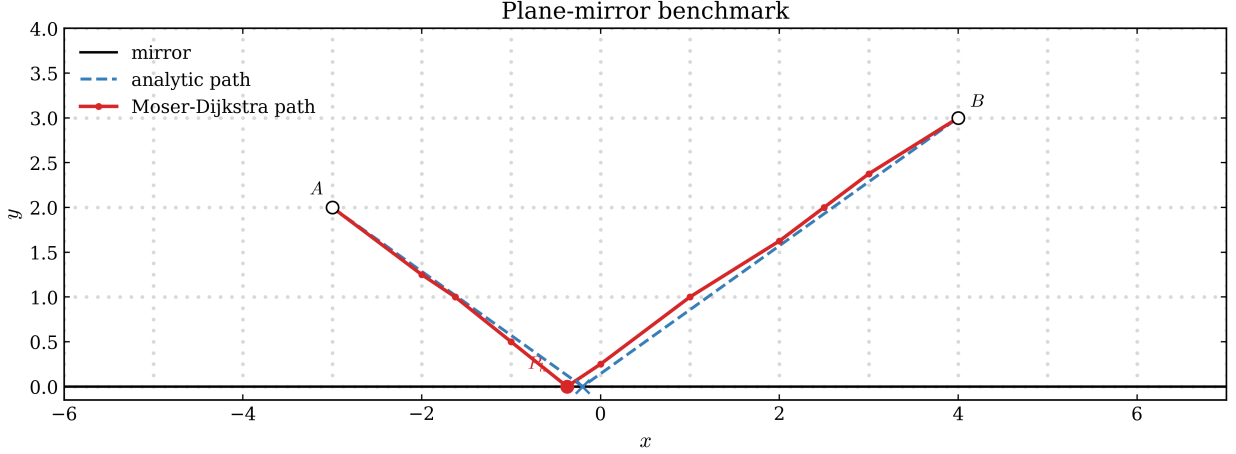


Figure 1: Baseline geometry. Gray points show the Moser boundary sampling, the dashed blue line is the analytic reflected ray, and the red polyline is the path recovered by the graph solver. The numerical and exact reflection contacts are visibly distinct.

3.3 Exactly one mirror contact

Let Γ_h be the set of graph nodes on the mirror. Two Dijkstra runs produce distance fields $d_A(v)$ and $d_B(v)$. During both runs, every node in Γ_h is absorbing: its distance is recorded, but its outgoing edges are not relaxed. Therefore, a branch reaching a candidate mirror point cannot have touched or travelled along the mirror earlier.

The constrained discrete Fermat functional is

$$F_h(P) = d_A(P) + d_B(P), \quad P \in \Gamma_h, \quad (9)$$

and the numerical contact is

$$P_h = \arg \min_{P \in \Gamma_h} F_h(P). \quad (10)$$

The complete path is reconstructed by joining the predecessor chains from A to P_h and from P_h to B . The implementation checks that the path contains exactly one mirror node and that its reconstructed cost agrees with $F_h(P_h)$. This is a reusable mandatory-surface constraint layered on the common graph solver, not a separate ray formula for the mirror problem.

4 Validation protocol

A single optical-path error is insufficient because a shallow minimum can give a nearly correct functional value at a visibly displaced path. We therefore record the following independent quantities:

- signed and relative optical-path errors;
- contact displacement $\|P_h - P_{\text{exact}}\|$;
- macroscopic angular residual, measured from A , P_h , and B ;
- local angular residual, measured from the actual two graph edges incident on P_h ;
- an approximate symmetric Hausdorff distance between the analytic and numerical polylines;
- the number of mirror contacts and graph-complexity statistics.

Because all graph weights are exact positive Euclidean lengths in this test, the discrete constrained minimum should satisfy

$$\mathcal{L}_h \geq \mathcal{L}_{\text{exact}} \quad (11)$$

up to floating-point roundoff.

Two experiments are performed. First, a nine-run parameter sweep independently varies

$$h \in \{1, 0.5, 0.25\}, \quad q \in \{1, 3, 7\}. \quad (12)$$

Second, a bias ensemble samples six exact reflection angles from 10° to 60° and eight horizontal translations over one main-cell period. Each (h, q) pair then contains 48 geometries. Medians, 90th percentiles, and maxima are reported so that a favorably aligned individual case cannot determine the conclusion.

5 Results

5.1 Baseline case

Table 1 summarizes the baseline calculation with $h = 1$ and $q = 7$. The earlier prototype stored 25,792 per-cell chords. Global deduplication removes 3,132 repeated undirected edges without changing the physical result.

Table 1: Baseline numerical result. Lengths are in the dimensionless units of the benchmark geometry.

Quantity	Value
Main-cell size h	1
Interior points per side q	7
Graph nodes	917
Unique graph edges	22,660
Duplicate chords removed	3,132
Numerical contact P_h	$(-0.375, 0)$
Exact contact P_{exact}	$(-0.200, 0)$
Contact error	0.175
Numerical optical length $\mathcal{L}_h$	8.610664161
Exact optical length $\mathcal{L}_{\text{exact}}$	8.602325267
Relative optical-path error	9.6938×10^{-4}
Macroscopic angle residual	2.865°
Local terminal-edge residual	4.970°
Approximate path Hausdorff distance	0.116
Mirror contacts	1

The relative optical-path error is only 0.0969%, yet the local reflection law is violated by nearly 5° . Thus, agreement in $\mathcal{L}$ alone would give an overly optimistic assessment of the numerical ray.

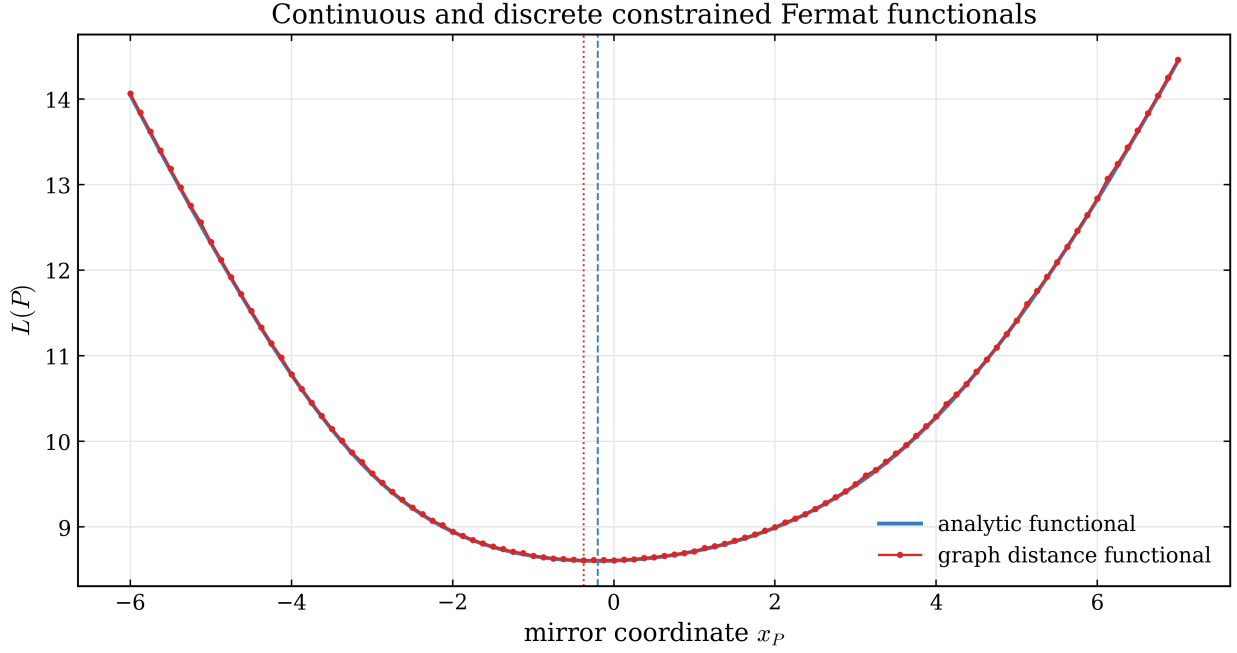


Figure 2: Continuous constrained functional from Eq. (3) and the graph functional from Eq. (9). The dashed blue and dotted red vertical lines mark the exact and numerical minimizers. Their functional values are close because the minimum is shallow.

5.2 Independent spatial and angular refinement

For this scale-aligned geometry, the measured relative optical-path error is numerically unchanged when h is reduced at fixed q (Table 2). Refining the main cells repeats the same finite local direction set; it does not eliminate the associated anisotropic graph metric. Increasing q lowers this directional error floor.

Table 2: Relative optical-path error in the nine-run sweep. Each value in a row was reproduced for $h = 1$, 0.5, and 0.25 in this particular aligned geometry.

q	Relative optical-path error
1	1.3072×10^{-2}
3	2.9749×10^{-3}
7	9.6938×10^{-4}

Contact and angular errors are not monotone in either resolution parameter. An exact mirror node or a symmetric pair of graph directions can occur by accidental commensurability. Consequently, isolated zero angular residuals are not evidence of asymptotic convergence.

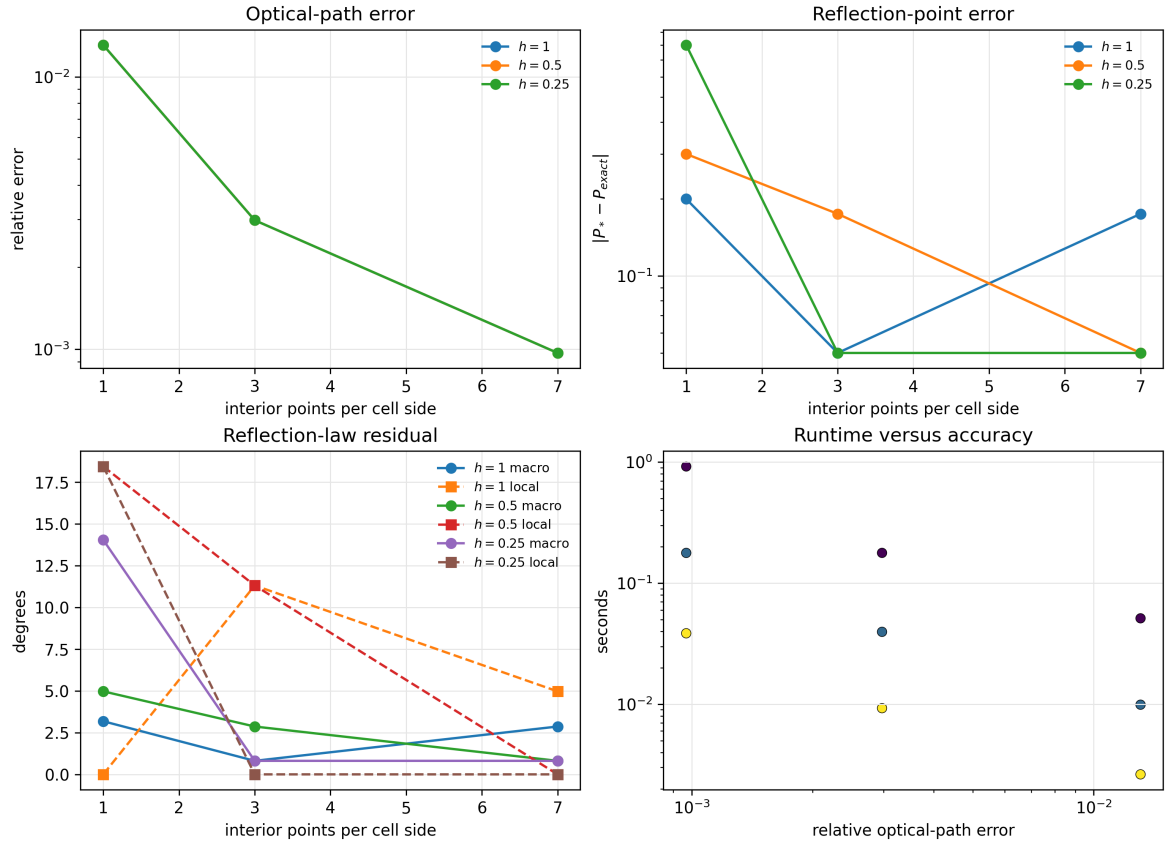


Figure 3: Nine-run resolution sweep. Optical-length, contact, macroscopic angular, and local terminal-edge errors are shown separately. The plots demonstrate that different observables need not converge at the same rate or monotonically.

5.3 Angle-and-phase ensemble

Table 3 reports the most resolved tested direction set, $q = 7$. Halving h improves the typical contact and ray-geometry errors, but the optical-path error remains near the same q -dependent floor. The two parameters therefore control distinct discretization effects.

Table 3: Ensemble statistics for $q = 7$ over 48 angles and phase shifts per value of h .

Metric	$h = 1$		$h = 0.5$	
	Median	90th pct.	Median	90th pct.
Absolute relative $\mathcal{L}$ error	9.02×10^{-4}	1.57×10^{-3}	9.25×10^{-4}	1.66×10^{-3}
Contact error	0.0600	0.1050	0.0338	0.0730
Macroscopic angle residual	1.434°	3.034°	0.858°	1.883°
Local angle residual	2.424°	6.911°	1.143°	5.440°
Hausdorff distance	0.0732	0.0953	0.0482	0.0748

At $q = 1$, the worst contact error reaches one full length unit for $h = 1$, and the 90th-percentile macroscopic angular residual exceeds 12° . A path that looks plausible on a low-direction graph is therefore not necessarily a quantitatively valid optical ray.

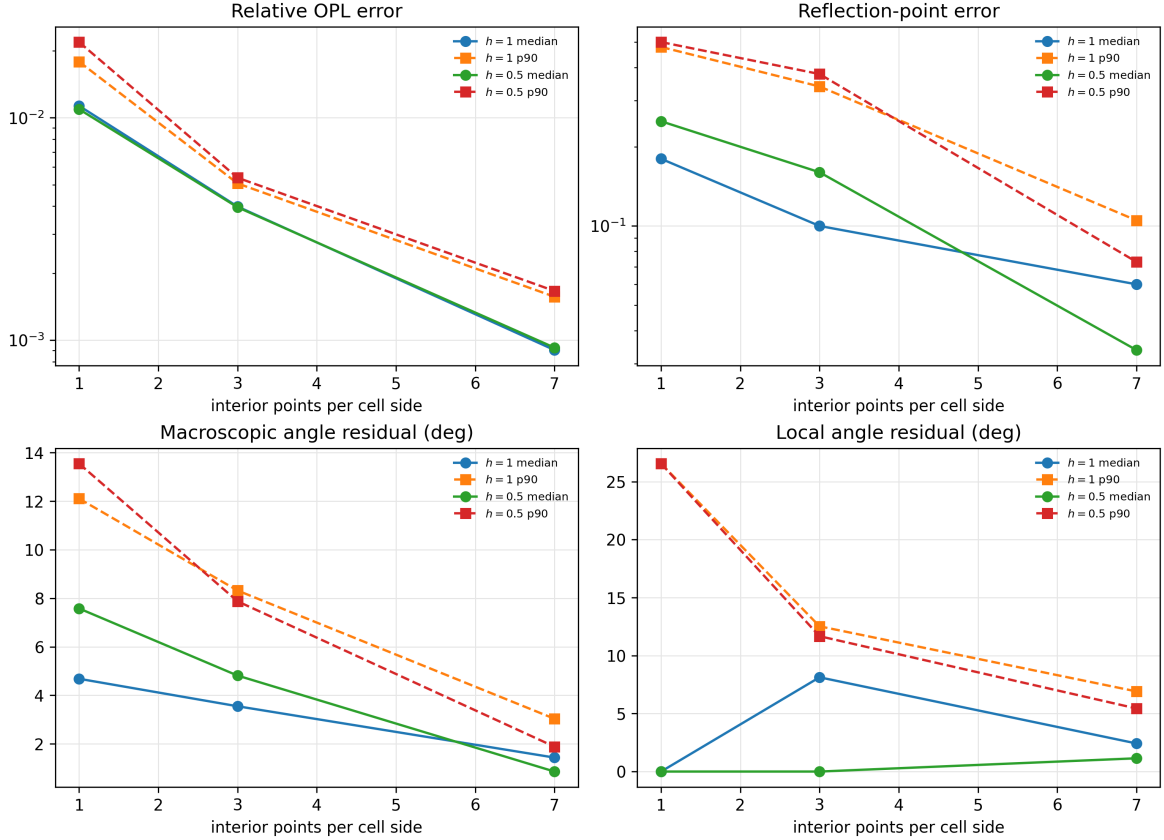


Figure 4: Distributions from the angle-and-phase ensemble. Increasing q primarily reduces the directional optical-length floor, while decreasing h improves typical spatial geometry once the direction set is sufficiently rich.

6 Flat-interface Snell benchmark

6.1 Continuous reference and reuse of the graph solver

Two homogeneous half-planes meet at $\Gamma = \{(x, 0) : x \in \mathbb{R}\}$. The upper and lower refractive indices are n_1 and n_2 , respectively. For a source $A = (x_A, y_A)$ above the interface, a receiver $B = (x_B, y_B)$ below it, and a contact $P = (x_P, 0)$, the continuous optical length is

$$\mathcal{L}(P) = n_1 \sqrt{(x_P - x_A)^2 + y_A^2} + n_2 \sqrt{(x_B - x_P)^2 + y_B^2}. \quad (13)$$

Differentiating with respect to the interface coordinate gives

$$\frac{d\mathcal{L}}{dx_P} = n_1 \sin \theta_1 - n_2 \sin \theta_2. \quad (14)$$

Therefore, the stationary minimum satisfies

$$n_1 \sin \theta_1 = n_2 \sin \theta_2. \quad (15)$$

The continuous reference contact is found independently by bisection of $d\mathcal{L}/dx_P = 0$ and is never supplied to the graph solver.

The lattice, edge deduplication, off-grid endpoint insertion, Dijkstra implementation, and mandatory-surface operator are unchanged from the mirror benchmark. If an edge crosses the material interface at Q , its weight is integrated exactly as

$$w_{ij} = n_1 \|\mathbf{r}_i - Q\|_2 + n_2 \|Q - \mathbf{r}_j\|_2. \quad (16)$$

Interface nodes are absorbing in the two distance-field calculations, and the minimum of Eq. (9) selects one crossing. Thus, Snell’s law is not inserted into the numerical path construction; it is used only to validate the resulting ray.

The baseline configuration is

$$A = (-3, 2), \quad B = (4, -3), \quad n_1 = 1, \quad n_2 = 1.5. \quad (17)$$

The interface coincides with a main-lattice row, so the optical bend occurs at an explicit graph node. Section 7 removes this alignment assumption by inserting intersection nodes and splitting every cut cell.

6.2 Baseline numerical result

Table 4 reports the result for $h = 1$ and $q = 7$. The graph contains 1,729 nodes and 44,852 unique edges; global deduplication removes 6,732 repeated chords.

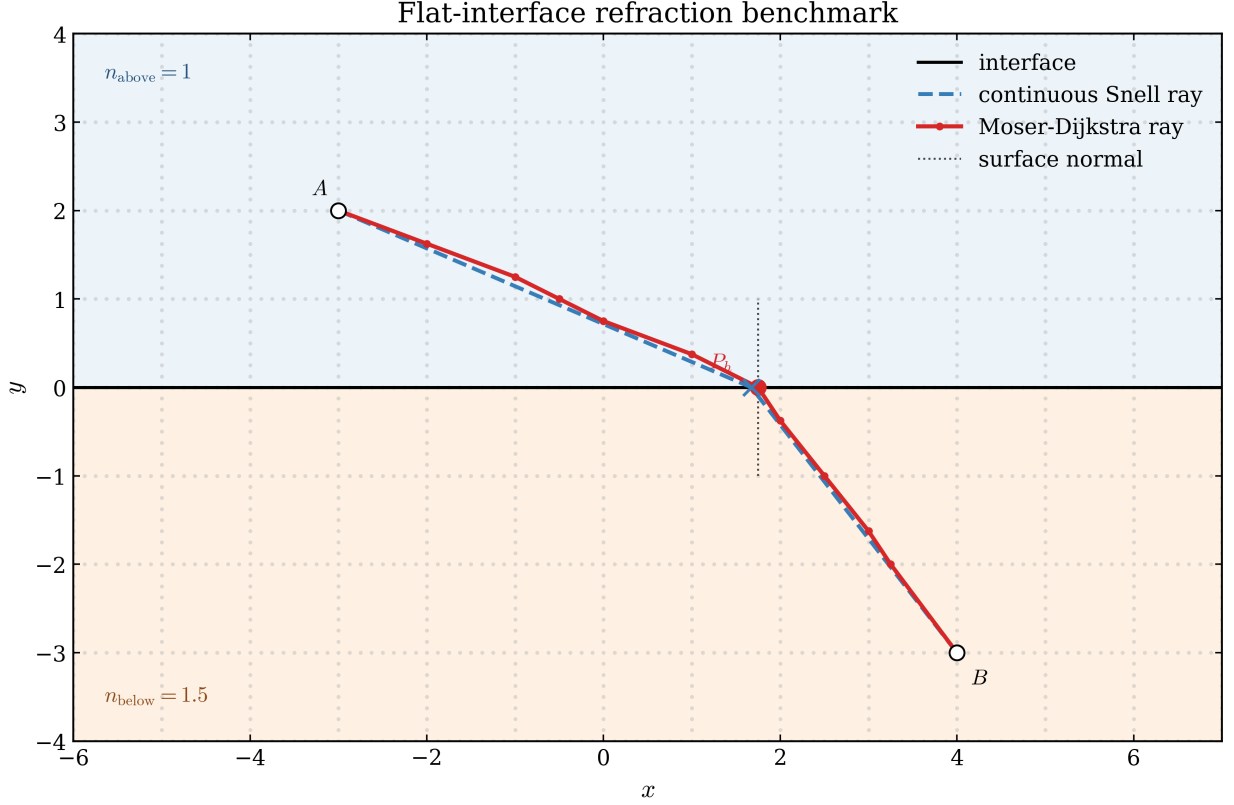


Figure 5: Baseline flat-interface refraction. The dashed blue curve is the continuous Snell ray, while the red polyline is recovered by the unchanged Moser–Dijkstra engine. The numerical interface contact is marked by the red point and the continuous contact by the blue cross.

Table 4: Baseline flat-interface refraction result. Angles are measured from the interface normal.

Quantity	Value
Continuous contact $P_{\text{continuous}}$	(1.673046239, 0)
Numerical contact P_h	(1.750000000, 0)
Contact error	0.0769538
Continuous optical length	10.778053561
Numerical optical length	10.788814202
Relative optical-path error	9.9838×10^{-4}
Exact angles (θ_1, θ_2)	$(66.8297^\circ, 37.7990^\circ)$
Numerical macro angles (θ_1, θ_2)	$(67.1663^\circ, 36.8699^\circ)$
Normalized macroscopic Snell residual	1.4424×10^{-2}
Normalized local Snell residual	4.1585×10^{-2}
Approximate path Hausdorff distance	0.09742
Interface contacts	1

To retain the tangential direction rather than compare only acute angles, the reported normalized residual is

$$\varepsilon_S = \frac{|n_1(\hat{\mathbf{s}}_{\text{in}} \cdot \hat{\mathbf{t}}) - n_2(\hat{\mathbf{s}}_{\text{out}} \cdot \hat{\mathbf{t}})|}{\max(n_1, n_2)}, \quad (18)$$

where $\hat{\mathbf{t}}$ is tangent to the interface. The sign is preserved inside the difference, so a ray travelling in the wrong tangential direction cannot pass the test by matching angle magnitudes alone.

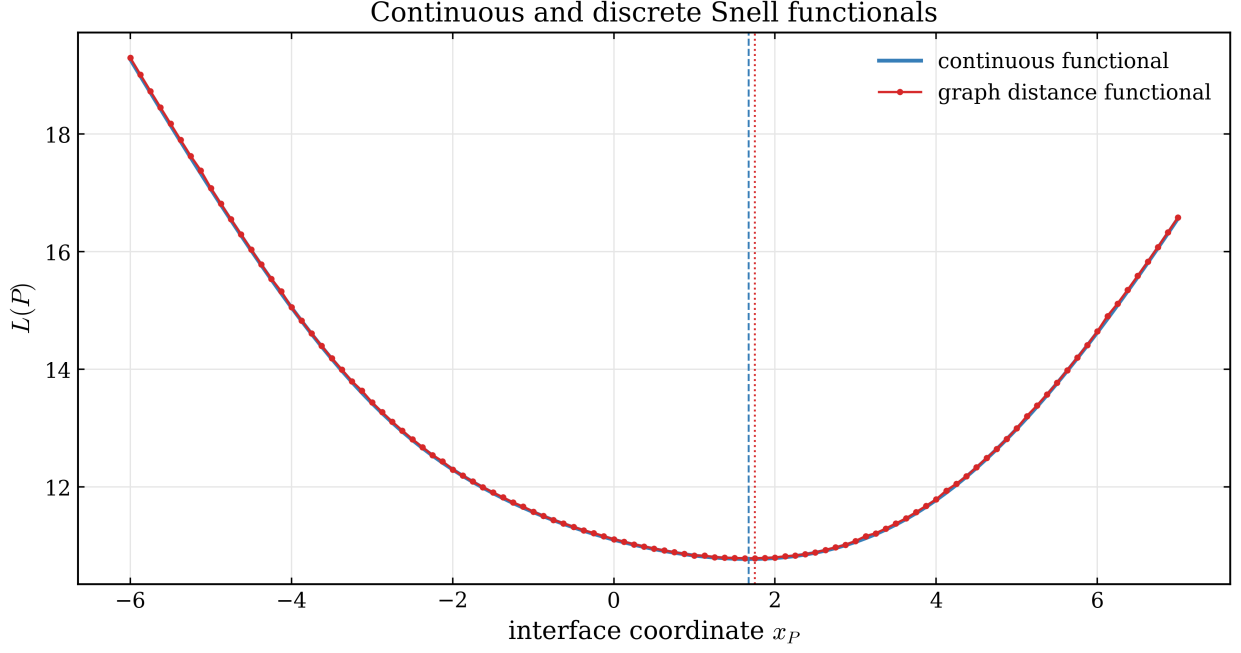


Figure 6: Continuous constrained functional from Eq. (13) and the discrete distance-field functional. The blue and red vertical lines mark the continuous and numerical contacts. Their functional values are close despite the visible contact displacement.

6.3 Separating spatial and angular refinement

The full nine-run sweep is shown in Table 5. Rows vary the angular parameter q , while columns vary the spatial scale h . This layout is important: comparing rows measures the effect of q , whereas comparing columns within one row measures the effect of h .

Table 5: Relative optical-path error for every tested (h, q) pair in the aligned baseline geometry.

q	Main-cell size		
	$h = 1$	$h = 0.5$	$h = 0.25$
1	5.3853×10^{-3}	5.3853×10^{-3}	5.3853×10^{-3}
3	3.4531×10^{-3}	3.5108×10^{-3}	3.5108×10^{-3}
7	9.9838×10^{-4}	9.2292×10^{-4}	9.2292×10^{-4}

At fixed q , reducing h leaves the baseline optical-path error nearly unchanged because the same finite local direction set is repeated on smaller cells. Increasing q changes that direction set and substantially lowers the directional optical-metric floor. Spatial refinement still improves local geometry: at $q = 7$, the normalized local Snell residual falls from 4.16×10^{-2} at $h = 1$ to 3.72×10^{-3} at $h = 0.25$. The selected contact remains $x_h = 1.75$ throughout this particular aligned case; finer interface sampling alone does not overcome the total directional bias accumulated in both half-planes.

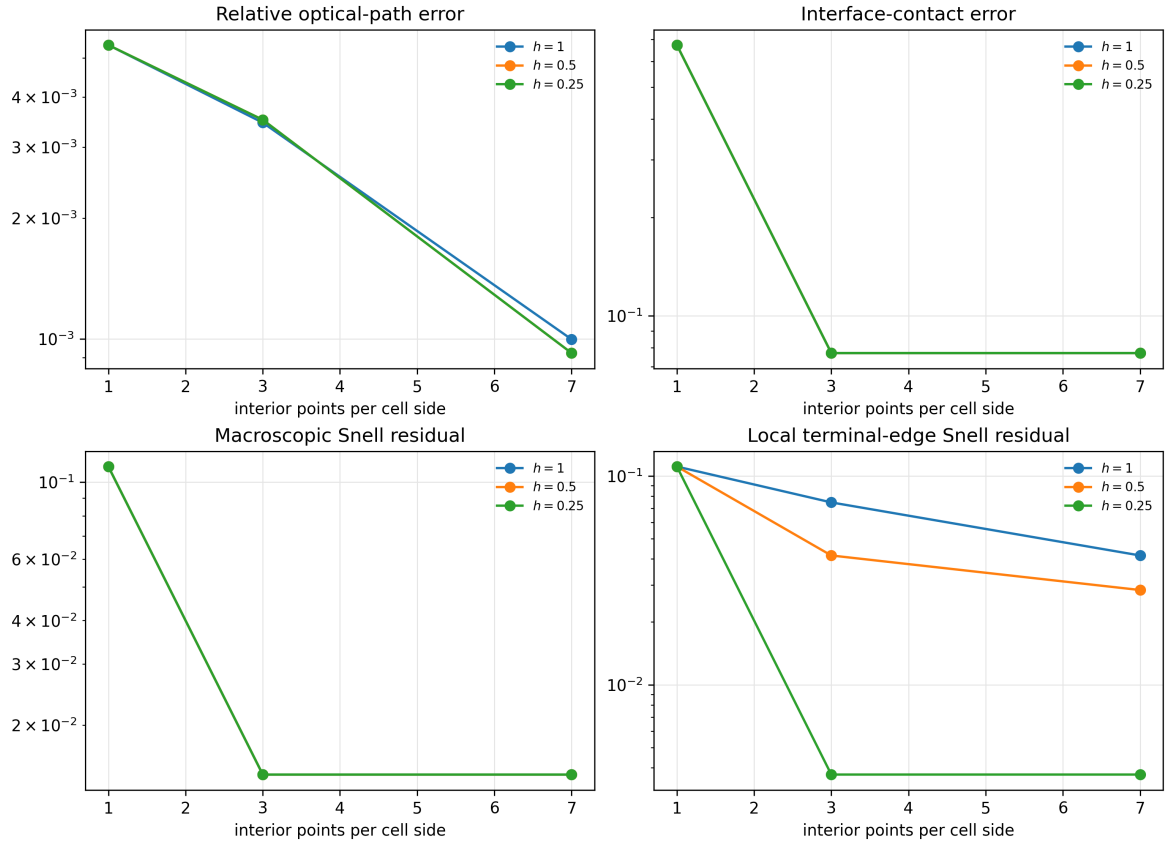


Figure 7: Nine-run refraction sweep. Curves with different h often overlap in optical-path and macroscopic metrics, while the local terminal-edge residual can improve under spatial refinement. Increasing q produces the dominant reduction in the directional error floor.

6.4 Angle, contrast, and phase ensemble

The refraction ensemble combines three index ratios, four exact incidence angles, eight contact phases, two cell sizes, and three values of q , for 576 runs in total. Each (h, q) summary contains 96 rays. Table 6 shows the most directionally resolved cases, $q = 7$.

Table 6: Refraction ensemble statistics for $q = 7$.

Metric	$h = 1$		$h = 0.5$	
	Median	90th pct.	Median	90th pct.
Absolute relative $\mathcal{L}$ error	7.57×10^{-4}	1.36×10^{-3}	5.51×10^{-4}	1.10×10^{-3}
Contact error	0.0450	0.0950	0.0338	0.0875
Normalized macro Snell residual	0.0190	0.0428	0.0140	0.0350
Normalized local Snell residual	0.0242	0.0745	0.0242	0.0745
Macro angle error	0.858°	1.754°	0.647°	1.531°
Hausdorff distance	0.0601	0.0890	0.0425	0.0848

Halving h improves the typical optical-path, contact, macroscopic-law, and path-shape errors. The local residual percentiles do not improve at fixed q , consistent with a repeated finite local direction set. By contrast, at $h = 1$ the median optical-path error decreases from 1.15% for $q = 1$ to 0.0757% for $q = 7$, and the 90th-percentile macroscopic angle error decreases from 9.84° to 1.75° . A quantitative Snell ray therefore requires angular enrichment in addition to spatial refinement.

As a stress check, four former $q = 7$ worst cases were rerun at $h = 0.25$ and $q = 15$. Their relative optical-path errors lie between 1.98×10^{-4} and 2.54×10^{-4} ; every path still has exactly one contact. A separate 6×6 sweep of independent source and receiver vertical phases gives a worst relative optical-path error below 9.6×10^{-4} for both $h = 1$ and $h = 0.5$. Halving h halves the contact quantization and improves the macroscopic Snell residual, while the direction of an individual terminal edge remains non-monotone at fixed q .

7 Non-grid-aligned straight interface

7.1 Interface-conforming cut cells

Let the material interface be the oriented line

$$\Gamma = \{\mathbf{r} : (\mathbf{r} - \mathbf{r}_0) \cdot \hat{\mathbf{n}} = 0\}, \quad (19)$$

with tangent $\hat{\mathbf{t}}$. For every main cell that has corners on both sides of Γ , the builder computes the two exact line–rectangle intersections. The resulting cut segment receives q interior Moser points, just like an ordinary cell side. Cell nodes are divided into positive and negative convex subsets; interface nodes belong to both, and each subset is completed separately.

This construction enforces a topological invariant: no graph edge joins two points strictly on opposite sides of Γ . Every transmitted graph path must therefore meet an explicit interface node at which it can bend. The same absorbing-surface operator then enforces exactly one crossing. For any residual edge whose endpoint lies on the interface, the half-plane is selected from the other endpoint; for a transverse edge the exact intersection weight in Eq. (16) is used after replacing the horizontal split by the oriented line.

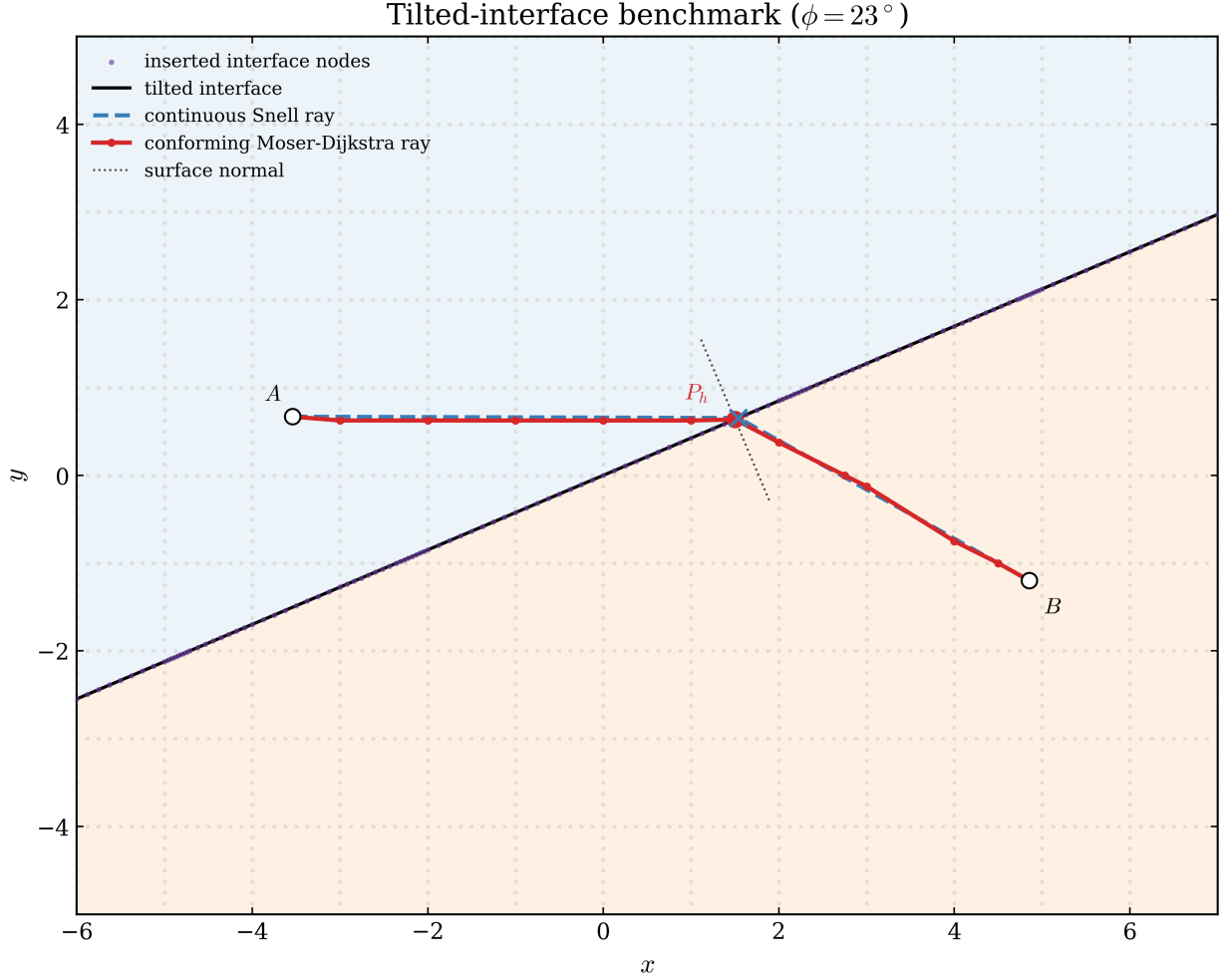


Figure 8: Baseline non-grid-aligned refraction at $\phi = 23^\circ$. Purple points are the interface nodes inserted into the cut cells. The dashed blue ray is the rotated continuous solution and the red polyline is the path recovered by the common Moser–Dijkstra solver.

7.2 Rotated invariant benchmark

The interface angle is $\phi = 23^\circ$. In local tangent–normal coordinates, the endpoints and indices are identical to the horizontal Snell case:

$$A' = (-3, 2), \quad B' = (4, -3), \quad n_+ = 1, \quad n_- = 1.5. \quad (20)$$

A rigid rotation cannot change the continuous optical length or local Snell angles. Hence the Stage 2 continuous solution is an invariant external benchmark, whereas the graph remains fixed in global Cartesian coordinates and genuinely experiences oblique cuts.

Table 7 reports the $h = 1, q = 7$ result. Seventeen main cells are cut, producing 137 interface nodes. The graph contains 2,273 nodes and 58,197 unique edges.

Table 7: Baseline tilted-interface result.

Quantity	Value
Continuous contact	(1.540047, 0.653711)
Numerical contact	(1.500000, 0.636712)
Contact error	0.04351
Continuous optical length	10.778053561
Numerical optical length	10.784982020
Relative optical-path error	6.4283×10^{-4}
Exact angles (θ_1, θ_2)	$(66.8297^\circ, 37.7990^\circ)$
Numerical macro angles (θ_1, θ_2)	$(66.6353^\circ, 38.3141^\circ)$
Normalized macroscopic Snell residual	7.9734×10^{-3}
Normalized local Snell residual	1.4742×10^{-2}
Approximate path Hausdorff distance	0.04220
Interface contacts	1
Cut cells / interface nodes	17/137

7.3 Orientation and phase ensemble

The tilted-interface ensemble combines two cell sizes, two angular resolutions, three index ratios, four interface angles, and eight two-dimensional translations, for 384 runs. Each (h, q) row in Table 8 summarizes 96 rays. Every ray obeys the continuous optical lower bound and contains exactly one interface contact.

Table 8: Median and 90th-percentile errors in the tilted-interface ensemble.

h	q	Relative $\mathcal{L}$	Contact	Macro Snell	Local Snell
1.0	3	.00399/.00703	.132/.325	.0246/.0561	.0835/.185
1.0	7	.00113/.00265	.0757/.179	.0127/.0303	.0382/.0925
0.5	3	.00371/.00602	.136/.338	.0256/.0744	.0637/.188
0.5	7	.000917/.00148	.0798/.191	.0157/.0365	.0370/.0788

Increasing q clearly improves every typical metric. Spatial refinement from $h = 1$ to $h = 0.5$ improves the optical functional modestly but does not monotonically improve absolute contact or macroscopic errors across the rotated ensemble. This is not a missed-interface artifact: the cut-cell invariant holds in every run. Rather, finite chord directions still dominate some geometries, and changing h changes the phase at which an oblique line cuts the Cartesian cells.

For the 23° baseline, raising the angular resolution to $q = 15$ reduces the relative optical-path error to 2.77×10^{-4} at $h = 1$ and 2.55×10^{-4} at $h = 0.5$. The contact error becomes 0.02439 and the macroscopic residual 0.00453 for both cell sizes. The Hausdorff error falls from 0.03567 to 0.01927 when h is halved, while the single-terminal-edge residual remains non-monotone. This again separates whole-ray convergence from the orientation of one final graph chord.

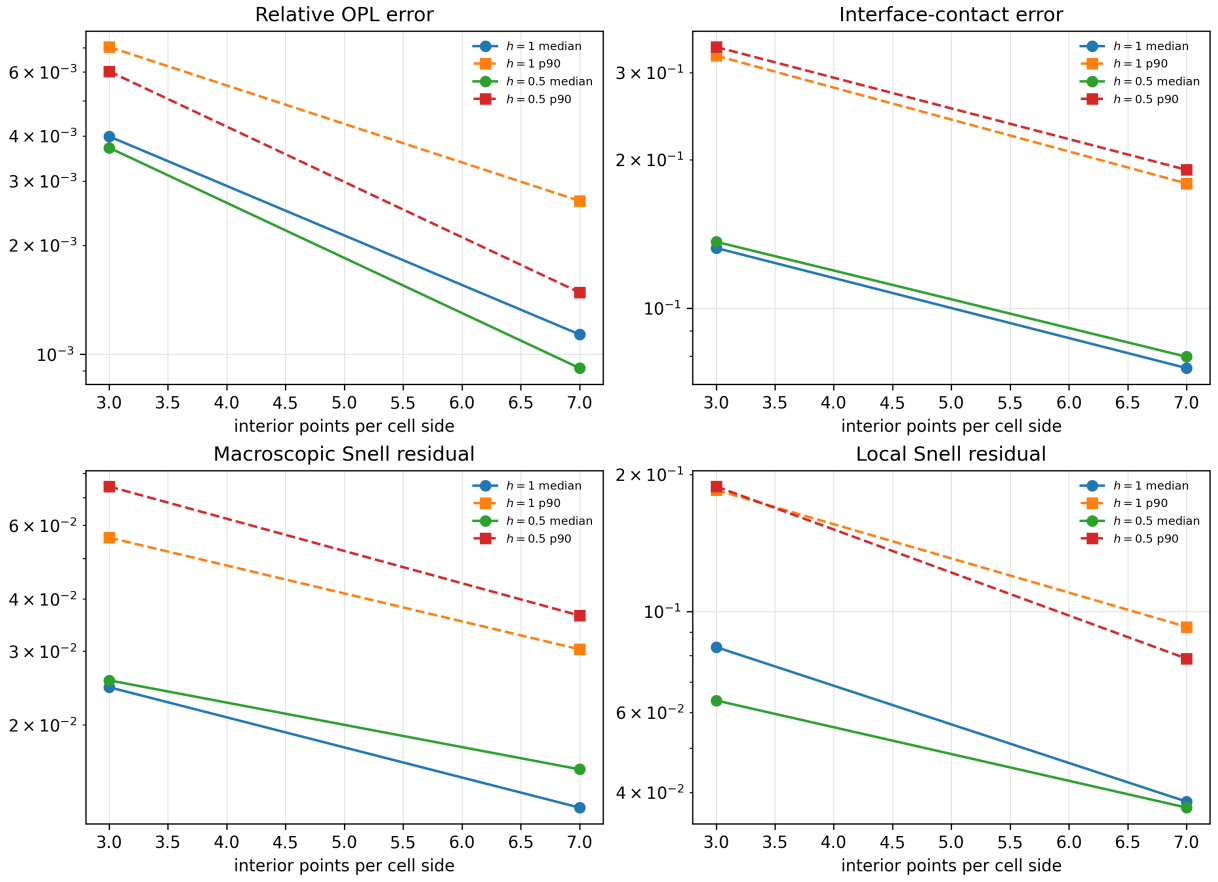


Figure 9: Tilted-interface ensemble. Increasing q from 3 to 7 improves the typical optical-path, contact, macroscopic, and local errors. Halving h at fixed q does not force monotone contact convergence because the finite chord-direction set and the phase of each oblique cut remain active.

8 Finite triangular prism

8.1 Two unknown contacts

The prism is a finite triangular glass region with exterior index $n_0 = 1$ and interior index $n_g = 1.5$. The prescribed physical sequence is air–glass–air. Its entry and exit contacts, P_1 and P_2 , are both unknown. The continuous constrained functional is

$$\mathcal{L}(P_1, P_2) = n_0 \|A - P_1\| + n_g \|P_1 - P_2\| + n_0 \|P_2 - B\|. \quad (21)$$

The baseline endpoints are

$$A = (-5, 1), \quad B = (5, 0.3), \quad (22)$$

and the prism vertices are

$$(-2, 3 - 2\sqrt{3}), \quad (0, 3), \quad (2, 3 - 2\sqrt{3}). \quad (23)$$

The stationarity of Eq. (21) with respect to motion of each contact along its face yields two independent Snell conditions,

$$n_0 \sin \theta_{1,0} = n_g \sin \theta_{1,g}, \quad n_g \sin \theta_{2,g} = n_0 \sin \theta_{2,0}. \quad (24)$$

Neither equation is supplied to the graph solver.

8.2 Staged multi-source Dijkstra formulation

Running a separate glass-region shortest-path calculation for every candidate P_1 would be correct but wasteful. Instead, the first propagation records the entry labels

$$C_1(P_1) = d_A(P_1). \quad (25)$$

Every reachable entry node is then inserted into one priority queue with its own nonzero label. A single multi-source propagation through the glass gives

$$C_2(P_2) = \min_{P_1} [C_1(P_1) + d_g(P_1, P_2)]. \quad (26)$$

A backward air-region field from B completes the exit functional,

$$F_h(P_2) = C_2(P_2) + d_B(P_2). \quad (27)$$

The winning multi-source label retains its root, so minimization of Eq. (27) recovers both P_1 and P_2 . This jointly optimizes the contacts and is not a greedy choice of the cheapest entry node.

The same unit-weight geometric Moser graph is reused in all three stages; its edge weights are multiplied by the index of the active material. Propagation is restricted successively to the exterior entry half-plane, the strict interior of the triangular prism, and the exterior exit half-plane. These regions are convex, so every chord whose endpoints survive a stage mask stays inside that material. Entry labels are fixed to prevent propagation along the first face, exit nodes are absorbing, and the base and prism vertices are excluded from the glass stage. Thus the recovered topology is one entry, one exit, and no base contact.

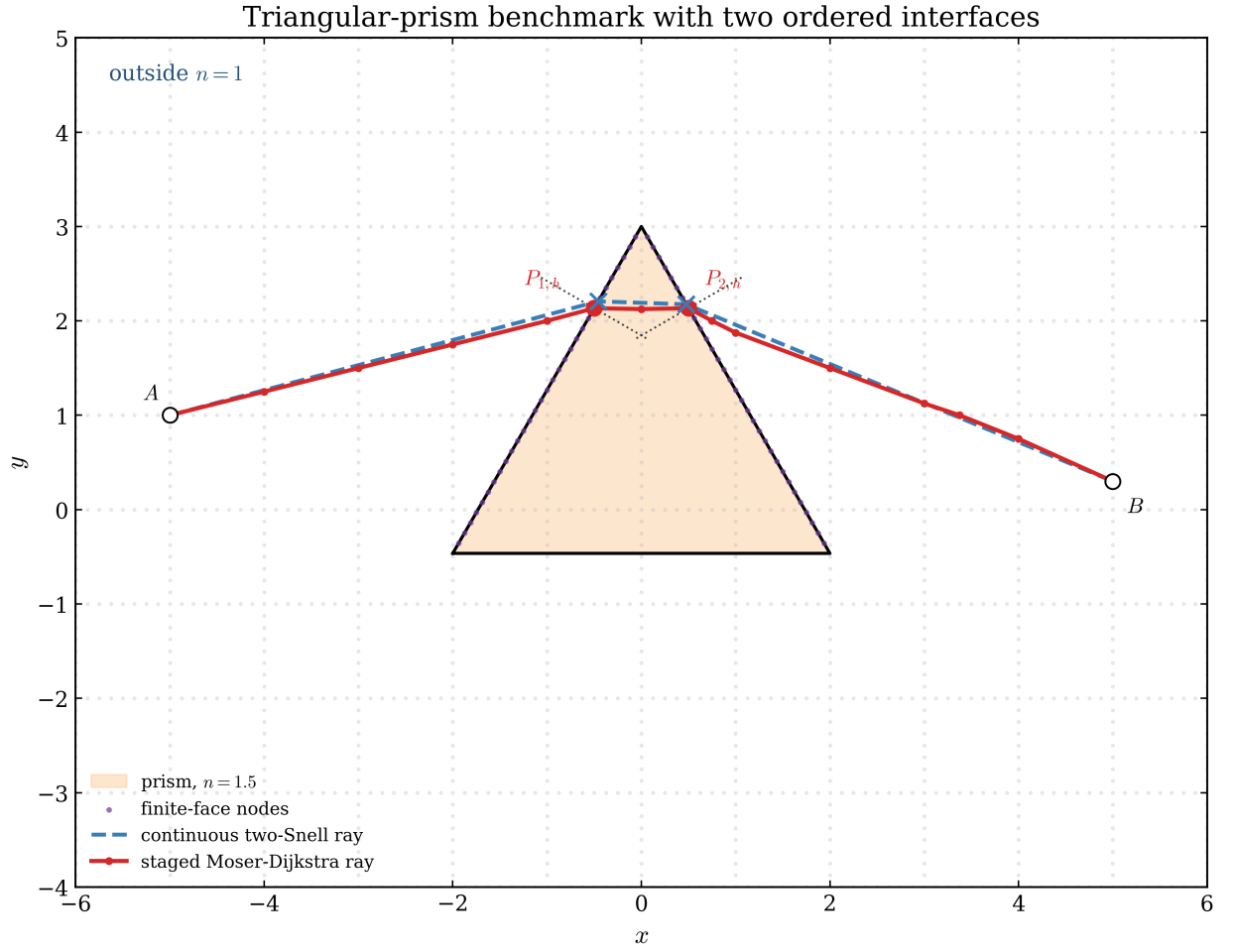


Figure 10: Finite triangular-prism baseline. Purple points discretize the two active faces. The dashed blue polyline is the independent continuous minimum of Eq. (21); the red polyline is reconstructed from the three staged distance fields.

8.3 Continuous reference and baseline result

An independent two-variable optimizer parameterizes both finite faces and alternates one-dimensional bisections of the two convex coordinate subproblems. It returns

$$P_{1,*} = (-0.457916, 2.206865), \quad (28)$$

$$P_{2,*} = (0.476171, 2.175248), \quad (29)$$

$$\mathcal{L}_* = 10.9987188725. \quad (30)$$

The signed tangential-momentum residuals of both continuous Snell conditions are below 8×10^{-15} .

For $h = 1$ and $q = 7$, the graph contains 1,852 nodes and 48,837 unique edges. Each active face contains 31 candidate nodes. Tables 9 and 10 separate the two interface errors from the whole-path functional.

Table 9: Baseline errors at the two prism faces. Angles are incidence and refraction angles measured from the local normal.

Quantity	Entry face	Exit face
Continuous contact	$(-0.457916, 2.206865)$	$(0.476171, 2.175248)$
Numerical contact	$(-0.500000, 2.133975)$	$(0.500000, 2.133975)$
Contact error	0.08417	0.04766
Exact angles	$(44.8801^\circ, 28.0614^\circ)$	$(31.9386^\circ, 52.5153^\circ)$
Macro angles	$(44.1437^\circ, 30.0000^\circ)$	$(30.0000^\circ, 52.1733^\circ)$
Normalized macro Snell residual	0.03569	0.02658
Normalized local Snell residual	0.01297	0.08214

Table 10: Whole-path triangular-prism baseline.

Quantity	Value
Continuous optical length	10.9987188725
Numerical optical length	11.0051463469
Relative optical-path error	5.8438×10^{-4}
Approximate path Hausdorff distance	0.07677
Entry / exit / base contacts	1/1/0
Contact order	entry before exit
Path nodes	15
Graph nodes / unique edges	1,852/48,837

The exit local residual is larger than the relative functional error because it measures one terminal graph chord. The macroscopic ray, contact positions, and total optical length are substantially more accurate. The distinction is visible in Fig. 10.

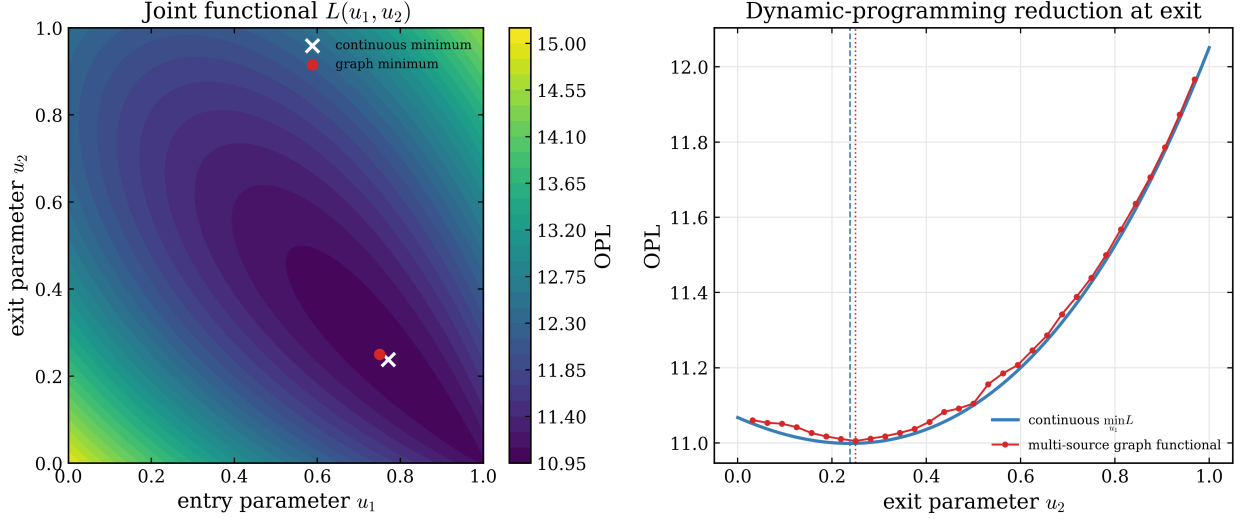


Figure 11: Left: the joint continuous functional over the two normalized face parameters. Right: minimization over the entry parameter reduces the problem to an exit functional. Red samples are produced by the multi-source graph propagation in Eq. (26).

8.4 Preliminary resolution matrix

Table 11 varies h and q for the fixed baseline geometry. All six paths remain above the continuous lower bound, retain the correct contact order, and avoid the base.

Table 11: Preliminary triangular-prism resolution matrix. Each residual is the larger value from the two active faces.

h	q	Relative $\mathcal{L}$	Contact	Macro Snell	Local Snell	Hausdorff
1.0	3	.004636	.2977	.1418	.2126	.2712
1.0	7	.000584	.0842	.0357	.0821	.0768
1.0	15	.000324	.0842	.0357	.0495	.0736
0.5	3	.002208	.0842	.0357	.0130	.1587
0.5	7	.000588	.0842	.0357	.0495	.0736
0.5	15	.000244	.0529	.00941	.0178	.0607

The strongest tested case, $h = 0.5$, $q = 15$, reduces the relative optical-path error to 2.44×10^{-4} and the maximum normalized macroscopic Snell residual to 9.41×10^{-3} . The non-monotone repetitions at fixed q are consistent with the directional and phase effects already observed for one interface. This matrix is not yet a prism ensemble over source position, apex angle, grid phase, and refractive index.

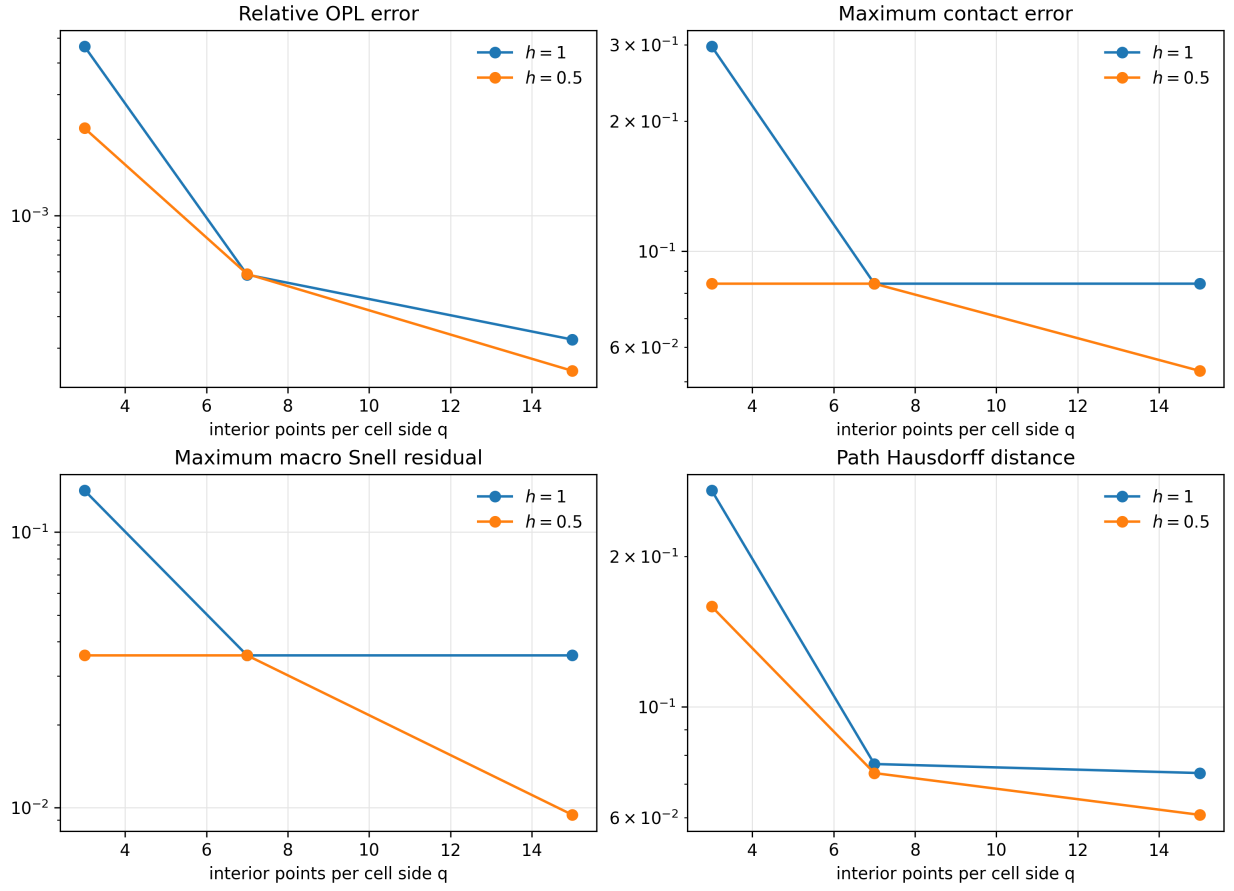


Figure 12: Resolution dependence for the fixed triangular-prism geometry. Raising q gives the dominant improvement; simultaneous spatial and angular refinement produces the best tested contact and Snell metrics.

9 Discussion and limitations

The four benchmarks support ten conclusions.

1. The common graph solver reproduces minimizing reflection and refraction without inserting either analytic ray into the algorithm.
2. Absorbing surface nodes enforce first contact and prevent an unphysical path from travelling along the zero-thickness surface.
3. Exact piecewise optical weights preserve a sharp material discontinuity without altering Dijkstra’s algorithm.
4. The recovered transmitted ray satisfies Snell’s law to the measured discretization accuracy even though Eq. (15) is not used during path construction.
5. Spatial scale h and angular richness q must be varied independently.
6. Functional convergence is weaker evidence than simultaneous convergence of contact position, ray shape, and local optical laws.
7. Interface-conforming cut cells remove the possibility that a graph edge skips a non-aligned straight discontinuity.
8. Removing this topological error does not remove finite-direction anisotropy; angular enrichment remains necessary after cutting cells.
9. A single multi-source propagation with nonzero entry labels jointly optimizes two unknown prism contacts without one solve per entry node.
10. Ordered convex-region masks generalize first-contact absorption to a finite air–glass–air sequence while excluding the unused prism base.

The result has a precise scope. Positive-weight Dijkstra finds a global minimum, whereas the full Fermat principle concerns stationary paths and can include maxima, saddles, multiple arrivals, and virtual rays. The present tests validate only global-minimum branches. Later cylindrical-mirror work must examine the complete discrete surface functional $F_h(P)$ for stationary points rather than select only its global minimum.

The present refractive geometry now supports arbitrarily oriented straight interfaces and one prescribed pair of finite prism faces, but a curved surface requires a curve–cell intersection routine, ordered surface segments, and consistent local normals. The prism solver does not yet choose among competing face sequences. That extension can be represented by a layered state graph while preserving the same distance-field engine. The refraction ensembles also cover only transmission from lower to higher refractive index; total internal reflection, Fresnel amplitudes, polarization, interference, and diffraction are outside the model.

10 Conclusion

The aligned, non-aligned, and finite-face benchmarks demonstrate a functioning universal Moser-lattice and Dijkstra formulation for constrained optical rays. The same graph and shortest-path architecture recovers reflection and Snell refraction through optical-length minimization and agrees with all four exact baseline optical lengths to below 0.1%. The tilted case shows that a sharp material interface can be made graph-conforming, while the prism shows that two unknown contacts can be combined efficiently through multi-source dynamic programming. More importantly, the

expanded metrics reveal that geometrical and local-law errors can remain much larger than the functional error suggests. This fixes the validation standard for later curved systems: every stage must report topology, ensemble-based geometrical and angular errors, and optical length separately.

A Reproducibility

From the project root, the benchmark and experiments are generated by

```
python scripts/run_plane_mirror.py
python experiments/plane_mirror_convergence.py
python experiments/plane_mirror_bias.py
python scripts/run_flat_refraction.py
python experiments/flat_refraction_convergence.py
python experiments/flat_refraction_bias.py
python experiments/flat_refraction_stage2_5.py
python scripts/run_tilted_refraction.py
python experiments/tilted_refraction_bias.py
python scripts/run_prism.py
python experiments/prism_convergence.py
PYTHONPATH=src python -m unittest discover -s tests -v
```

The problem module defines only the physical configuration and observables. Graph construction, edge integration, shortest paths, and mandatory-surface constraints remain in shared modules so that later benchmarks cannot replace the universal framework with private ad hoc solvers. The complete validation suite contains 35 passing tests.

References

- [1] Thanaporn Arunthong, Laddawan Rianthakool, Khanchai Prasanai, Chakrit Na Takuathung, Sakchai Chomkokard, Wiwat Wongkokua, and Noparit Jinuntuya. Numerical solutions to the variational problems by dijkstra’s path-finding algorithm. *Applied Sciences*, 14(22):10674, 2024.